

Tianyi Zhang

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RESEARCH INTERESTS

Trustworthy AI systems, end to end. I study proxy fidelity across the ML pipeline: whether training data reflects the real distribution, learned objectives capture intended behavior, and evaluation metrics track genuine quality and safety. My work spans data and benchmark integrity, representation learning, retrieval evaluation, reward/specification gaming, and monitor robustness.

EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science

Atlanta, GA

Aug 2025 - Dec 2026

Emory University

Bachelor of Science in Quantitative Science (Honors)

Atlanta, GA

Aug 2021 - May 2025

PROFESSIONAL EXPERIENCE

LinkedIn

AI/ML Engineer Intern | Feed AI, LLM Retrieval

Mountain View, CA

May 2026 - Present

- Expanded retrieval evaluation beyond Recall by building and production-scale validating a backward-compatible **PySpark scorecard** over **2.7B member-candidate pairs**; combined ranking quality, diversity, exposure, lifecycle, eligibility, and efficiency signals to support more reliable, member-value-aware model selection.
- Established a matched lifecycle evaluation controlling a **30x population imbalance**; exposed **+25.1% Recall@10** and **+26.8% NDCG@10** but **21.6% lower ILD** for Semi-cold versus matched Active members. Prompt analysis identified median histories of 31 versus 135 activities as a representation confounder.
- Built a backend-agnostic **Adaptive User Representation Layer** that decouples user understanding from candidate generation. Its compact user-state contract captures short- and long-term intents, lifecycle, confidence/freshness, and evidence provenance for ANN, GR, SID, or larger-model consumers; implemented the first producer with profile/history multi-tower Qwen3 encoding.
- Improved distributed training and model-serving reliability through layered configuration, dry-run validation, reproducible manifests, execution tracking, searchable run catalogs, train/serve parity checks, and serving observability while preserving legacy workflow compatibility; defined one-axis-at-a-time validation across representation, model capacity, retrieval backend, and lifecycle policy.

NVIDIA - Georgia Tech Efficient & Intelligent Computing Lab

Student Researcher | LLM Evaluation, Agentic AI

Atlanta, GA

Jul 2025 - Dec 2025

- Co-authored **AgentSuite (ICML 2026)**, a component-based pipeline for auditing LLM-agent benchmarks; surfaced flaws affecting ~23% of tasks across 6 benchmarks and shifting ~63% of model rankings.
- Built taxonomy-guided LLM-judge and structural-detector stages, matching expert labels at **0.79-0.87 F1** and surfacing 100+ undocumented issues.

Emory Graph Mining Lab

Undergraduate Researcher | Graph Learning, AI for Science

Atlanta, GA

Dec 2023 - May 2025

- Built a graph-neural-network pipeline over ~8K patients and first-authored a Granger-directed brain-connectivity method at **AIME 2025**.
- Engineered a Transformer-based clinical time-series pipeline modeling transitions across latent functional-connectivity states; oral presentation at **IEEE EMBC 2025**.

PUBLICATIONS

H. Suh*, S. Lee*, B. Ji*, R. Khare, B. Khan, H. Kim, **T. Zhang**, et al. "AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline." *International Conference on Machine Learning (ICML)* 2026.

T. Zhang, K. Han, J. Nie, C. You, S. van Rooij, J. Stevens, B. Dunlop, C. Gillespie, C. Yang. "Causal Brain Connectivity: Integrating Granger Directed Graphs in fMRI Analysis." *International Conference on Artificial Intelligence in Medicine (AIME)* 2025. (First Author)

J. Nie, K. Han, **T. Zhang**, C. You, S. van Rooij, J. Stevens, B. Dunlop, C. Gillespie, C. Yang. "Brain Network State Transformer: Leveraging State Functional Connectivity." *IEEE Engineering in Medicine and Biology Society (EMBC)* 2025. (Oral)

SELECTED PROJECTS

CoAct: Governance Substrate for Multi-Agent Coding

- Building a dependency-free Go governance substrate that lets coding agents work safely in one repository through capability policies, advisory locks, worktree isolation, merge gates, a shared task board, and an append-only journal for replayable oversight.

Agent-Loop Observability

- Developing observability tools for diagnosing repeated tool calls, context loss, unrecalled memory writes, cost runaway, and safety-monitor failures in long-running LLM-agent trajectories.

OPEN SOURCE

NVIDIA NeMo RL - Contributor to NVIDIA's scalable reinforcement-learning and LLM post-training library, including checkpointing and response-dataset improvements.

TECHNICAL SKILLS

Programming: Python, Go, SQL, Java, R, C/C++

ML/AI: PyTorch, PyTorch Lightning, CUDA, Hugging Face Transformers, vLLM, SGLang, MLflow

Data/Systems: Spark/PySpark, FSDP, Docker, AWS, GitHub Actions, Linux, PostgreSQL, MongoDB, Redis